

Boolean Logic

Truth Table

Is a table which represents all the possible values of logical variables/statements along with all the possible results of the given combinations of values.

X	Y	Result
1	1	1
1	0	0
0	1	0
0	0	0

Logical Operators

- NOT Operator
- OR Operator
- AND Operator

NOT Operator

- Unary Operator
- Operation Name - Complementation
- Denoted by - (bar) or ' symbol
- Example : $\overline{X}$
- Means complement of X

NOT Operator

- Truth Table for NOT Operator

X	$\overline{X}$
0	1
1	0

OR Operator

- Binary Operator
- Operation Name - Logical Addition
- Denoted by + symbol
- Example : $X + Y$
- Means X OR Y

OR Operator

- Truth Table for OR Operator

X	Y	$X + Y$
0	0	0
0	1	1
1	0	1
1	1	1

AND Operator

- Binary Operator
- Operation Name - Logical Multiplication
- Denoted by . symbol
- Example : $X . Y$
- Means X AND Y

AND Operator

- Truth Table for AND Operator

X	Y	X . Y
0	0	0
0	1	0
1	0	0
1	1	1

LOGIC GATES

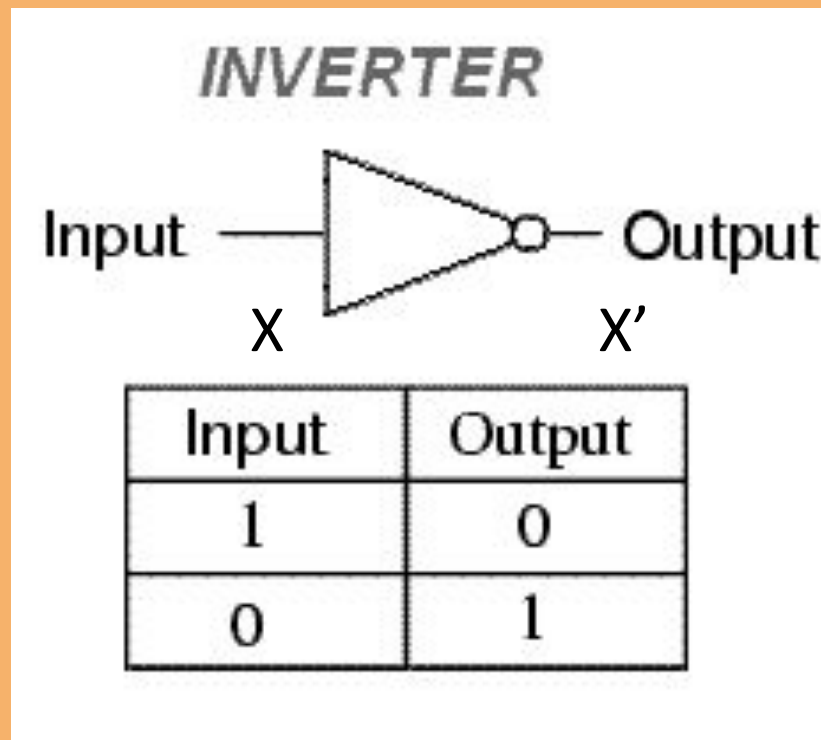
- A Gate is simply an electronic circuit which operates on one or more signals to produce an output signal.
- Three types of Gates are:
 - Inverter(NOT gate)
 - OR gate
 - AND gate

Inverter (NOT) Gate

- One input signal and one output signal only.
- Output state is always the opposite of the input state.
- Output is sometimes called the complement of the input.

Inverter (NOT) Gate

- Symbol and Truth table of NOT gate:-



OR Gate

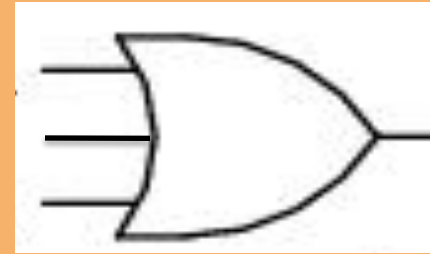
- Two or more input signals and but only one output signal.
- If any of the input signals is 1, the output signal is 1.

OR Gate

- Symbols:-



- 2 INPUT



- 3 INPUT

OR Gate

- Truth Tables:-

- 2 INPUT

X	Y	R
0	0	0
0	1	1
1	0	1
1	1	1

$$R = X + Y$$

- 3 INPUT

X	Y	Z	R
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

$$R = X + Y + Z$$

AND Gate

- Two or more input signals and but only one output signal.
- When all the inputs are 1 then the output is 1 otherwise output is 0 only.

AND Gate

- Symbols:-



- 2 INPUT



- 3 INPUT

AND Gate

- Truth Tables:-

- 2 INPUT

X	Y	R
0	0	0
0	1	0
1	0	0
1	1	1

$$R = X \cdot Y$$

- 3 INPUT

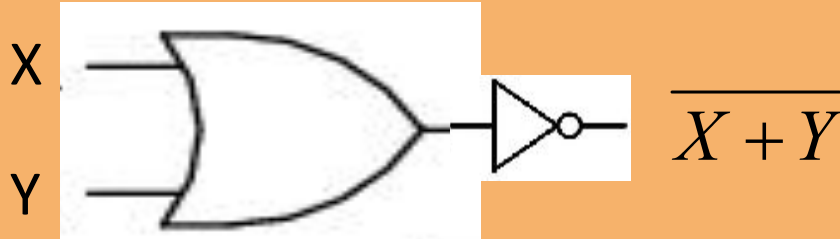
X	Y	Z	R
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

$$R = X \cdot Y \cdot Z$$

DeMorgan's First Theorem

- It states that

- $$\overline{X + Y} = \overline{X} \overline{Y}$$



DeMorgan's First Theorem

- Truth Table of

- $\overline{X + Y} = \overline{X} \overline{Y}$

X	Y	$\overline{X}$	$\overline{Y}$	$X + Y$	$\overline{X + Y}$	$\overline{X} \overline{Y}$
0	0	1	1	0	1	1
0	1	1	0	1	0	0
1	0	0	1	1	0	0
1	1	0	0	1	0	0

DeMorgan's Second Theorem

- It states that

- $$\overline{XY} = \overline{X} + \overline{Y}$$



DeMorgan's Second Theorem

- Truth Table of

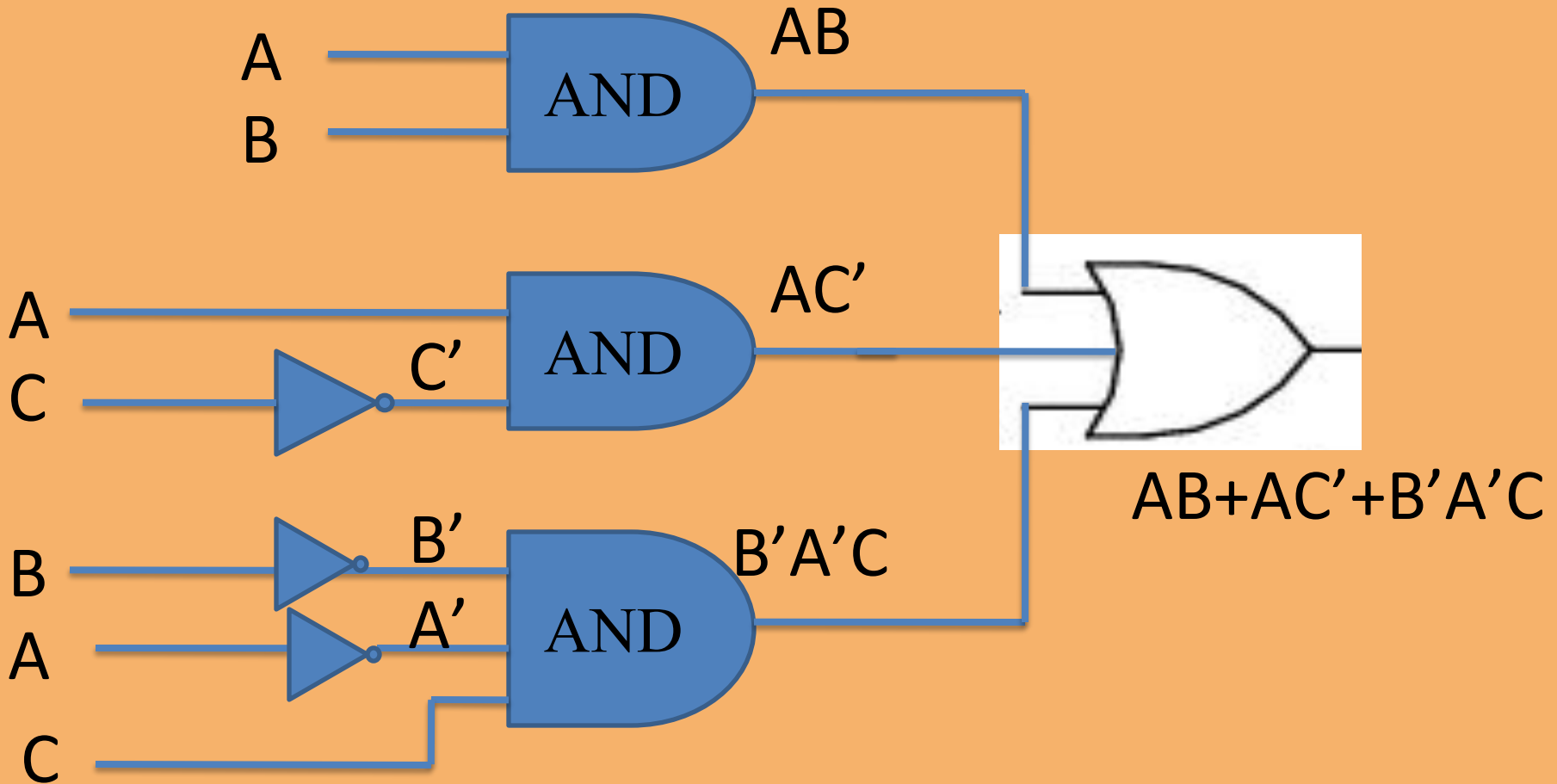
- $$\overline{XY} = \overline{X + Y}$$

X	Y	$\overline{X}$	$\overline{Y}$	$\overline{XY}$	$X + Y$	$\overline{X + Y}$
0	0	1	1	1	0	1
0	1	1	0	0	1	0
1	0	0	1	0	1	0
1	1	0	0	0	1	0

Circuit Design

- Design a circuit to realise the following:
- $F(a,b,c) = AB + AC' + B'A'C$
- (A AND B) OR (A AND (NOT C)) OR ((NOT B) AND (NOT A) AND C)

Circuit Design



END